Wastewater as domestic resource infrastructure

National nutrients, sludge energy, costs and import-equivalent supply. Evidence reviewed October 16-17, 2026. Water reuse sales excluded. Analysis date follows America/Chicago, October 16, 2026.

Finding: municipal wastewater can provide a meaningful recurring nutrient supply and distributed energy. The strongest project candidates combine recoverable products with genuinely avoided treatment and disposal costs. Gross potential is not additional production or profit.

50% screening case	Annual output / value	Interpretation
Phosphorus / ammonium N / K2O	300,000 / 610,700 / 424,413 metric tonnes	Gross nutrient scenario; existing reuse not fully subtracted
P/N/K replacement value	$2.38 billion	Dated retail-equivalent nutrient value; before costs
AD combined heat and power	5.36 TWh net electricity + 3.61 TWh usable heat	$588M energy value at stated bill-offset proxies
AD upgraded gas alternative	47.9 billion cubic feet	$147M Henry Hub gross value; before upgrading costs
HTL liquid fuel alternative	262.5M gallons diesel + 87.5M other fuel	$1.28B diesel-only wholesale value; conversion assumed

Do not add the three energy alternatives. They allocate the same underlying sludge carbon differently. The nutrient and energy values also use different price bases and counterfactuals; this report does not present a combined national profit.

What is established, and what is modeled

Commercial nutrient recovery and sludge electricity exist [S28-S29]. National inventories and technology studies provide starting points [S1-S9]. The 25/50/75% cases, costs, price realization, heat use and fuel-upgrading splits here are explicit screening assumptions. No measured 2026 fleet-output claim is made.

The most useful near-term sequence is to audit existing recovery, target concentrated sidestreams and flared gas, compare full treatment baselines, then match qualified products to actual buyers. A facility that is large by flow is not necessarily the best nutrient or financial opportunity.

Boundaries and data quality

Evidence layer	Basis	Limit
National flow	EPA approximately 34 billion gallons/day [S1]	Approximate reference, not a new national meter total
P inventory	2025 modeled CONUS sewage budget [S4]	Total approximately 0.60 Mt P inferred; existing reuse included
N/K concentrations	2009 EPA and 1996 NRC references [S2-S3]	40 mg/L total N, 65% ammonium; 15 mg/L K are assumptions
Sludge reporting	About 4 million dry metric tonnes from 2024 submissions [S9]	Reporting coverage and treated-solids boundary differ from raw-sludge studies
Energy potential	2013 NREL methane; 2020 PNNL HTL [S7-S8]	Dated gross technical/economic potential, not new current output
Facility count	2022 CWNS: 17,544 reported POTWs [S36]	Different universe from dashboard permit records
Dashboard inventory	34,665 facility-permit records; 5,333 selected outfalls	Overlapping registers, not unique-plant counts or additive flows

The dashboard discovery queue contains 14,948 EFF/ADC municipal permit records, with 1,545 heuristic flow-review flags. Application flow, design flow and annual DMR flow must stay separate [S37]. Flags prioritize source review and do not prove an error. The dashboard register is not used as a summed national flow denominator here.

Input sensitivity is wide: EPA concentration endpoints imply about 0.56-2.30 Mt/year ammonium-N before recovery; the older K concentration range implies 0.57-1.70 Mt K2O equivalent. These are sensitivity bounds, not confidence intervals. A national assay-weighted model is needed to narrow them.

Four distinct economic accounts

Replacement value: what comparable nutrients or energy cost a buyer. **Receipts:** what the recovery facility can actually sell. **Avoided baseline:** expenditure that genuinely disappears after retrofit. **Net incremental benefit:** receipts plus disjoint avoided expenses, minus new capital, operations and residual costs. Policy credits are a fifth, separate sensitivity.

Null values in the data package mean unestablished. They do not mean a resource has no value. Older composition references are marked as content proxies and are excluded from qualified-output valuations.

Annual resources and current value

Net recovery	P, tonnes	NH4-N, tonnes	K2O eq., tonnes	Gross replacement
25%	150,000	305,350	212,207	$1.19B
50%	300,000	610,700	424,413	$2.38B
75%	450,000	916,051	636,620	$3.57B

Recovery fractions include collection, extraction and qualification losses by assumption. They are not whole-plant recovery efficiencies established by a pilot. 75% is an ambitious sensitivity; dilute-potassium separation is particularly uncertain.

Nutrient	Reference value / metric tonne	50% case value
P	$3,951	$1.185B
N	$1,313	$802M
K2O equivalent	$922	$391M

DTN retail survey, October 5-9, published October 14: ammonia $977, DAP $934 and potash $502 per short ton [S6]. Convert 1 short ton = 0.90718474 metric tonnes. N price uses 82% N ammonia. Subtract the 18% N component from DAP before allocating its remaining value to 46% P2O5. Potash uses 60% K2O. This keeps each nutrient counted once.

Struvite is a product route, not an extra resource

If all 300,000 tonnes of recovered P in the midpoint were converted into pure struvite, product-equivalent mass would be 2.38 Mt/year, containing 135,664 tonnes N and requiring 235,409 tonnes Mg. The remaining standalone recovered N is 475,037 tonnes. Magnesium and other reagent purchases belong in process costs. P/N nutrient-equivalent value is about $574 per tonne of pure struvite; no premium or magnesium sales credit is assumed.

That conversion is a stoichiometric illustration. Total P may require release, purification or an alternative phosphate route. Magnesium from desal brine could be investigated as an input, but it is not automatically compatible or free.

Physical replacement and additional supply

50% case	Import comparison	Meaning
Ammonium N: 610,700 t	32.1% of ammonia imports as N	Excludes imports of other N fertilizers
Potassium: 424,413 t K2O	7.6% of potash imports	Qualified grade and agronomic demand needed
Gross P: 300,000 t	67.4% of assumed P in rock imports	Includes P already reused; not additional supply
Disposal-P subset: 65,500 t	14.7% of assumed P in rock imports	Narrower additional-recovery target, conditional

2025 USGS import estimates: ammonia 1.9 Mt as N; potash 5.6 Mt as K2O; phosphate rock 3.4 Mt gross rock [S5]. The rock conversion assumes 30% P2O5, giving 445,155 tonnes P. Actual imported grade and finished phosphate imports require separate accounting. Nutrient-content equivalence does not imply interchangeable industrial feedstock or equal crop response.

A focused phosphorus target

The largest-plant study lists 67,000 tonnes P to landfill, 34,000 to incineration and 30,000 to other disposal: 131,000 tonnes/year [S4]. Recovery of 25/50/75% gives 32,750/65,500/98,250 tonnes P and about $129M/$259M/$388M P-only replacement value. This subset is already inside the gross P total; the listed categories are not a complete national flow balance.

Midpoint annual output is equivalent to about 54 days of phosphate-rock import P under the assumed grade. That is annual production equivalence, not a stored reserve. Existing land application and other recovery must be retained in the baseline.

Interstate gap filling

The study identifies urban P concentrations and agricultural shortfalls in parts of the Midwest [S4]. A real route needs a named producer, seasonal qualified output, a named buyer, crop nutrient demand, product-specific delivered cost and the foreign or domestic purchase actually replaced. No particular interstate route is verified by this national screen.

Three alternatives, one carbon account

Deployment share	CHP net power	CHP usable heat	RNG alternative	HTL diesel alternative
25%	2.68 TWh	1.80 TWh	24.0 Bcf	131.2M gal
50%	5.36 TWh	3.61 TWh	47.9 Bcf	262.5M gal
75%	8.05 TWh	5.41 TWh	71.9 Bcf	393.7M gal

AD-CHP: anaerobic microbes produce biogas; clean it and generate electricity plus useful heat. The model uses methane HHV 55.5 GJ/t, 35% electric efficiency, 40% recoverable heat, 15% electric parasitics and 50% of recovered heat available after internal loads. These are assumptions, not measured national averages. Availability affects required generator capacity.

AD-RNG: reserve 15% of gas energy for process heat and recover 95% of the remainder during upgrading. Sellable gas is an alternative to burning it in CHP. Compression, siloxanes/H2S removal, connections and methane slip remain material costs.

HTL: PNNL literature provides a rounded 1 billion gallon/year biocrude reference [S8]. Here, 70% finished-fuel volume yield and a 75% diesel share are assumptions. At 50% deployment, 500M biocrude gallons become 262.5M diesel and 87.5M other fuel gallons. Hydrogen demand, product density, energy content and actual refinery yields need a process model [S34].

Midpoint energy pathway	Annual gross value / offset proxy	Import-volume comparison
CHP electricity + usable heat	$524M power + $64M heat = $588M	Net power equals 17.8% of gross electricity imports
RNG alternative	$147M at Henry Hub	1.5% of gross gas imports
HTL diesel alternative	$1.283B diesel-only wholesale value	10.8% of gross distillate imports

A 2014 federal roadmap separately estimated 5.6 TWh/year full WRRF electricity potential [S39], below this NREL-based energy conversion. Half of that reference is 2.8 TWh before harmonizing net/gross boundaries. Different scopes and assumptions prevent treating either as a precise forecast.

Import ratios use 2025 gross volumes [S15-S18]. Actual displacement is unknown. Electricity must match time/location and may replace domestic generation; the U.S. is a net gas exporter. Distillate includes products beyond road diesel. Finished-fuel diesel-only value excludes other products and all upgrading expenses.

Prices, capital and full-system cost

Benchmark	Value / observation period	Use
Industrial electricity [S10]	$97.70/MWh, July 2026	Avoided-purchase proxy; actual municipal tariff differs
Wholesale electricity [S11]	$52/MWh, 2026 forecast	Alternative export-value proxy
Henry Hub [S12]	$2.95/MMBtu, September 2026	RNG commodity benchmark
Industrial gas [S13]	$4.58/Mcf, July 2026	Heat displacement, 85% assumed boiler efficiency
Gulf ULSD [S14]	$4.886/gallon, September 2026	Qualified diesel wholesale comparison

For the CHP midpoint, about 801 MW of gross generator nameplate is required at 90% capacity factor. The following are **capital budget stress tests, not vendor estimates**. Annualize at 8% over 20 years; assume $0.02/gross kWh O&M. Values below exclude costs omitted from the budget and adjustment for existing recovery.

Budget case	Capital	Annual capital + O&M;	Energy-value balance
Existing-gas retrofit	$2.40B	$371M	$217M
Broader retrofit	$6.40B	$779M	-$190M
Major new train	$16.01B	$1.76B	-$1.17B

The lower budget is only a plausible screen where gas/digestion infrastructure already exists. New digestion, sludge preprocessing, gas storage, interconnection, standby equipment and difficult residual management can exceed these allowances. Generic CHP guidance [S22-S23] is not a full sludge-project quote. Already-used biogas also has an opportunity cost.

At wholesale rather than industrial purchase prices, midpoint electricity value falls from $524M to $279M; heat value is unchanged. Only genuinely avoided energy charges count. Annual generation does not establish backup capability: black-start, islanding protection, fuel continuity and critical-load tests are separate requirements.

Break-even tests and baseline savings

At full benchmark realization, maximum all-in product-funded costs are about $3.95/kg P, $1.31/kg N and $0.92/kg K2O. At an assumed 75% realization, those ceilings become $2.96/$0.99/$0.69. Those are component-only ceilings. Struvite N may add value to the P route if not credited elsewhere. Verified treatment/disposal savings can also raise the ceiling, without double credit.

Illustrative cost stress	P route, $/kg P	Standalone N, $/kg N	K route, $/kg K2O
Lower	2	1	0.50
Middle	4	2	1.50
Higher	8	4	3.00

These rates are **assumed sensitivities, not nationally estimated costs**. They must cover annualized capital, labor, reagents including Mg, energy, finishing, freight, testing, rejects and residual management per qualified output. The P route includes struvite N co-recovery; standalone N cost is applied only to the remaining N, avoiding a duplicate process charge.

50% output case	All-in cost	Assumed receipts	Before avoided costs
lower cost stress	$1.29B	$1.78B	$497M
middle cost stress	$2.79B	$1.78B	-$1.00B
higher cost stress	$5.57B	$1.78B	-$3.79B

The middle stress case needs about $1.00B/year of disjoint avoided expenses or other support to break even. This is a sensitivity result, not a claim that such savings exist. Existing product revenue and land-application benefits must not be credited again.

Sludge conversion cost evidence

PNNL 2024 HYPOWERs mature-plant case reports $3.00/gal biocrude minimum selling price with avoided incineration and fuel credits; pioneer-plant analysis is much less favorable [S19]. Prices and credit assumptions belong to that case and date. It does not establish a nationwide 2026 cost. First-plant engineering and aqueous-phase treatment studies [S20-S21] reinforce the need for site-specific budgets.

A prior national HTL model estimated $1.4B/year disposal savings at modeled full deployment [S33]. A purely linear half-deployment reference would be $700M/year in that older model. Do not append that to an economics case already crediting disposal, or assume savings scale linearly at every facility. A current costed national net-benefit estimate remains unestablished.

Full resource register: known content versus output

Resource	Illustrative content	Status
Organic nitrogen before treatment	657,677 t content proxy	Recovery/value unknown
Magnesium in effluent	939,539 t content proxy	Recovery/value unknown
Calcium in effluent	2,348,848 t content proxy	Recovery/value unknown
Sulfur in reported sludge	40,000 t content proxy	Recovery/value unknown
Iron in reported sludge	68,000 t content proxy	Recovery/value unknown
Aluminum in reported sludge	20,000 t content proxy	Recovery/value unknown
Zinc in reported sludge	4,800 t content proxy	Recovery/value unknown
Copper in reported sludge	3,000 t content proxy	Recovery/value unknown
Manganese in reported sludge	1,000 t content proxy	Recovery/value unknown

Mg/Ca: old secondary-effluent concentrations multiplied by approximate national flow. Sulfur/metals: old typical sludge composition multiplied by the approximately 4 Mt reported treated-solids subset [S3,S9]. These are **context proxies, not measured national inventories**, and do not enter recovery or revenue totals. No metal or sulfur price is multiplied by an unverified saleable yield.

Additional resource	Route / evidence need
Coagulant Fe/Al	Acid recovery/reuse is studied [S31]; chemical demand and P release can change economics.
Silver, gold, critical metals	Assays, speciation, separations and buyers; detection alone is not a mineable reserve [S27].
Phosphate in ash	Part of the original P account; metal removal and plant availability matter.
Biochar / activated carbon	Competes for sludge carbon; product quality and contaminants not established.
PHA / organic acids / hydrogen	Alternative carbon or external-energy routes; not added to full energy potentials.
Deuterium	Natural inventory about 1.58 Mt D/year in approximate flow at assumed 150 ppm; qualified recovery/value unknown [S32].

The earlier isotope model also gives about 7.83 Mt/year pure-D2O mass equivalent. This is an atom inventory, not separated heavy water, an attainable annual product or a market-sized demand forecast. No research-isotope or future fusion revenue is assigned.

Additional energy opportunities need a heat sink

Wastewater sensible heat is separate from sludge chemical energy. Sewage-source heat pumps have measured site examples [S30], but national accessible heat depends on collection, temperature, seasonal demand and treatment constraints.

Thermal sensitivity assumption / result	Value
Temperature extraction / accessible share	5 C / 10% of approximate flow
Heat-pump COP	4
Source heat recovered	27.31 TWh/year
Electricity required	9.10 TWh/year
Heat delivered	36.42 TWh/year
Avoided gas-heat proxy / power expense	$646M / $889M per year
Operating balance before capex and O&M	-$244M/year

This particular gas-displacement case does not pay for its electricity at the selected national proxies. Cheaper power, displaced oil/electric resistance heating, cooling demand or better COP can change the result [S38]. It is not evidence against all sewage heat pumps. Do not add heat credits where CHP already serves the same load, and preserve treatment/discharge temperature requirements.

Efficiency can be an earlier, cheaper resource

DOE describes low/no-cost packages with 5-7% facility-wide savings [S25]. Applying an illustrative 6% to its historical 30.2 TWh/year sector reference [S24] yields 1.81 TWh/year, or $177M at the electricity proxy before implementation costs. Reconcile with the same post-efficiency load baseline used for CHP; it is not automatically additive.

Raw biogas also contains CO2. An assumed 60/40 methane/CO2 mix implies about 2.14 Mt CO2/year in the midpoint gas stream. Purification, compression and a matched buyer are missing. Carbon already in the sludge must not be credited twice; no CO2 sales or carbon-credit value is counted.

Contaminants, residuals and product gates

A physically recovered resource becomes resilience supply only when usable, dependable and deliverable. The model does not certify fertilizer, soil amendments, fuel specifications, emissions or permits.

Stream / risk	Required accounting	Release gate
PFAS	Track aqueous, solids, air and concentrate destinations; capture is not destruction.	Validated technology-specific treatment and residual destination [S26]
Microplastics	Capture to filters/sludge creates a residual burden.	Demonstrated destruction or controlled management; no automatic soil application
Metals	Recovery and concentration alter speciation and reject hazards.	Qualified product assays and controlled residue disposal
Pathogens / pharmaceuticals	Route-dependent treatment and byproduct behavior.	End-use testing and applicable process requirements
Salt / sodium / crop response	Nutrient mass does not establish agronomic equivalence.	Buyer/crop specification, availability and application timing [S35]
Fuel / biogas impurities	H2S, siloxanes, heteroatoms, catalyst poisons and methane loss.	Cleanup, emissions controls, product certification and measured net energy

Thermal sludge processing is not automatically a PFAS-destruction guarantee. EPA 2026 guidance highlights process-specific performance, emissions and products of incomplete destruction [S26]. Old biosolids compliance or a process label cannot replace current product evidence.

Cost items that must stay in the final project boundary

Collection; dewatering and drying; acid/base/Mg and other reagents; heat/electricity/H2; cleanup and upgrading; product finishing; laboratory testing; storage; freight; labor; maintenance; backup equipment; financing; startup/ramp losses; off-gas and aqueous treatment; residual transport/disposal; monitoring and decommissioning. Avoided expenses must use the same baseline boundary.

Current evidence cannot close all these items nationally. The data package keeps the missing values explicit instead of assuming zero. No national construction recommendation or verified profit is claimed.

What to prioritize and what to measure next

First priority: concentrated P/N streams, gas currently flared, and existing digesters needing efficient CHP or cleanup. Existing infrastructure can make a retrofit more attractive than building a new train. Protect existing recovery as part of the supply base; pursue additional output separately.

Second priority: regional HTL or other sludge-processing hubs where measured sludge supply, disposal costs, refinery access, heat and qualified products support a complete case. Combine routes only with an explicit carbon, nutrient and contaminant balance.

Selective research: potassium concentration/recovery; coagulant recycling; Mg exchange with brine facilities; ash phosphate; specialized industrial metal streams; isotope separation. Do not extrapolate municipal averages into industrial or mine streams.

Evidence needed	Resilience decision enabled
Unique plant and process stream; monthly flow/assays	Reliable annual yield, uncertainty and no duplicate accounts
Existing output, flare, land use and disposal costs	Additional supply and incremental economics
Equipment, heat sinks, parasitics and downtime	Net energy and realistic capacity
Buyer product/grade, seasonal demand and freight	Deliverable supply and true domestic/foreign replacement
Gas/energy/fertilizer imports by destination and product	Route-level import substitution, not just national ratios
Black-start/islanding, fuel storage, spares and outage tests	Actual operational resilience and reserve duration

The biggest plants concentrate much of the resource, but the optimal shortlist also includes smaller sites with concentrated streams, high disposal costs, nearby farms/industry or vulnerable power supply. A resilience score should expose these inputs rather than hide them in a single weighted number.

Separate register extensions: food/beverage, dairy/manure, fertilizer manufacturing, landfill leachate, mining/metallurgy, power-plant cooling, brackish/desal and decentralized sanitation. Confirm whether each stream already enters a municipal account before combining inventories. National all-wastewater totals remain incomplete.

Model contract and limitations

The accompanying ZIP contains inputs.json, results.json, run_model.py, the report text and this PDF. Python 3 standard library is sufficient to reproduce the numerical results: run **python run_model.py** in the extracted folder. No account or network access is needed.

Automated checks cover finite values, nutrient allocation, energy conservation, fuel-product splits, and retention of unknown margins. Report tables are generated from the same result objects where practical. No facility assays or engineering claims are invented by the checks.

• 25/50/75 nutrient fractions are net qualified-output assumptions; energy fractions are deployment shares of dated gross literature potentials, not equivalent recovery technology efficiencies.

• No total annual profit is reported. Product replacement value, plant-gate receipts, avoided costs and policy credits are different accounts.

• AD-CHP, AD-RNG and full-sludge HTL are alternatives here. A hybrid requires a separate explicit carbon/mass balance.

• Struvite N is inside total recovered N; do not add it or charge separate nitrogen-recovery costs to it. Mg reagent costs belong in the P route.

• CHP parasitic electricity is deducted once; heat reserved for internal process loads is not externally sold. Do not value the methane again after converting it to heat and power.

• Heat pumps consume electricity; do not count delivered heat as recovered source energy alone.

• Existing land application, generation, heat, products and baseline costs must be subtracted for additionality. Avoided expense must not be duplicated across nutrient and sludge trains.

• Imported nutrient content, gross electricity/gas volumes and diesel volumes are physical comparisons, not verified monetary trade-deficit reductions. Net gas exports and electricity transmission constraints matter.

• Old sludge composition times partial reported dry mass is a context proxy. Unknown qualified recovery/value is null, never zero-filled.

• Water reuse volume and sales value remain excluded; thermal energy is a separate resource account.

• Contaminants capture, soil-use clearance and destruction are distinct. No autonomous approvals, soil safety or emissions clearance inferred.

• No speculative deuterium fuel sales, carbon-credit revenue or critical-metal revenue included.

Source retrieval took place October 16-17, 2026 UTC. The local analysis date is October 16, 2026 in America/Chicago. Market observations have their own stated dates; source publication date is not a common measurement year.

Source register 1

S1 - EPA wastewater nutrient sources

Approximate national flow, not a new metered total. Open primary source

S2 - EPA Nutrient Control Design Manual, 2009

Reference concentration ranges, not national assays. Open primary source

S3 - National Research Council, 1996, composition tables

Old reference concentrations; low-confidence proxy. Open primary source

S4 - Islam et al., 2025 phosphorus spatial budget

doi:10.1016/j.scitotenv.2025.180851; modeled CONUS budget, includes existing reuse. Open primary source

S5 - USGS Mineral Commodity Summaries 2026

2025 estimated import denominators; compare on elemental units. Open primary source

S6 - DTN original retail survey, published October 14, 2026

October 5-9 prices; retail replacement value, not plant-gate receipts. Open primary source

S7 - NREL Biogas Potential in the United States, August 2013

Dated gross methane potential, not additional 2026 output. Open primary source

S8 - Seiple et al., 2020 sludge HTL fleet model

doi:10.1016/j.jenvman.2020.110852; rounded biocrude potential. Open primary source

S9 - EPA sewage sludge reporting, 2024 submissions

Reporting subset; treated solids differ from untreated sludge feedstock. Open primary source

S10 - EIA electricity prices, July 2026

Industrial national average 9.77 cents/kWh; proxy only. Open primary source

S11 - EIA October 2026 electricity outlook

2026 forecast wholesale average $52/MWh, not an offtake quote. Open primary source

S12 - EIA Henry Hub spot prices, September 2026

Monthly spot benchmark $2.95/MMBtu. Open primary source

S13 - EIA industrial natural gas prices, July 2026

Delivered national average $4.58/Mcf, not a plant contract. Open primary source

Source register 2

S14 - EIA petroleum spot prices, September 2026

Gulf ULSD $4.886/gal; regional wholesale benchmark. Open primary source

S15 - EIA electricity imports and exports, 2025

30,086,248 MWh gross imports; network/time matching remains necessary. Open primary source

S16 - EIA gas imports, 2025

3,163.625 Bcf gross imports; U.S. net gas exporter. Open primary source

S17 - EIA distillate imports, 2025

57,989,000 barrels; not solely road diesel. Open primary source

S18 - EIA crude imports, 2025

Approximately 6.2 million barrels/day; volume context only. Open primary source

S19 - PNNL HYPOWERs preliminary economics, 2024

PNNL-36528; case-specific costs depend on policy credits, disposal and maturity. Open primary source

S20 - PNNL site-specific first-plant design, 2024

PNNL-37046; first-plant design differs from mature-plant studies. Open primary source

S21 - PNNL aqueous byproduct treatment, 2024

HTL aqueous carbon/nitrogen require treatment; not free disposal. Open primary source

S22 - EPA CHP market analysis, October 2011

Historical performance/cost context, not current full digester quotes. Open primary source

S23 - EPA CHP procurement guidance

Generic CHP installed ranges; excludes a complete new sludge train. Open primary source

S24 - DOE wastewater energy baseline

Historical sector electricity reference. Open primary source

S25 - DOE wastewater efficiency measures

Low/no-cost improvement reference, not independently additive to generation. Open primary source

S26 - EPA 2026 PFAS destruction/disposal guidance

Capture is not destruction; technology-specific validation required. Open primary source

Source register 3

S27 - EPA Targeted National Sewage Sludge Survey

2006-07 samples; maxima/detection statistics are not present national mass loads. Open primary source

S28 - Clean Water Services nutrient recovery

Operator-reported existing product, not additional supply. Open primary source

S29 - DC Water Blue Plains

Current operator description of existing energy recovery. Open primary source

S30 - PNNL/DOE sewage heat-pump measurement case

Site case, not national performance. Open primary source

S31 - Chakraborty et al., recovered iron coagulant, 2020

doi:10.1016/j.seppur.2019.115894; acid/reuse and nutrient interactions. Open primary source

S32 - DOE hydrogen isotope handbook

Natural abundance reference; no recovered isotope revenue. Open primary source

S33 - DOE sludge HTL national economic discussion, 2021

Prior fleet-model disposal savings; not an extra generic credit. Open primary source

S34 - PNNL HTL process development units

Biocrude upgrading removes trace metals and heteroatoms. Open primary source

S35 - EPA agricultural reuse resources

Crop/end-use quality requirements; water-sales value excluded. Open primary source

S36 - EPA Clean Watersheds Needs Survey 2022

17,544 reported existing POTWs; different from ECHO permit count. Open primary source

S37 - EPA wastewater flow definitions

Application, design and DMR flows differ; preserve periods. Open primary source

S39 - Federal Biogas Opportunities Roadmap, 2014

Alternative WRRF electricity-potential reference; source scopes differ. Open primary source

S38 - DOE heat-pump technology guidance

Useful heat and electricity cost determine heat-pump value. Open primary source